

Name of Student: Shreyasi Mishra	PRN: 25070122294
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 5	
TITLE : Collection Classes and StringBuffer	

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Programme

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

```
import java.util.ArrayList;
import java.util.Scanner;

public class ToDoList {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        ArrayList<String> tasks = new ArrayList<>();

        System.out.print("Enter the number of tasks: ");

        int n = sc.nextInt();

        sc.nextLine();

        for (int i = 0; i < n; i++) {

            System.out.print("Enter Task " + (i + 1) + ": ");

            tasks.add(sc.nextLine());
```

```

    }

    StringBuffer sb = new StringBuffer();

    sb.append("\n----- TO-DO LIST ----- \n");

    for (int i = 0; i < tasks.size(); i++) {

        sb.append(i + 1).append(" ").append(tasks.get(i)).append("\n");

    }

    System.out.println(sb);

    sc.close();

}

}

```

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```

import java.util.ArrayList;

import java.util.Scanner;

public class CourseRegistration {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        ArrayList<String> courses = new ArrayList<>();

        int choice;

        do {

            System.out.println("\n===== Student Course Registration System
=====");

            System.out.println("1. Add Course");

            System.out.println("2. Remove Course");

            System.out.println("3. View Registered Courses");

            System.out.println("4. Exit");

```

```
System.out.print("Enter your choice: ");

choice = sc.nextInt();

sc.nextLine();

switch (choice) {

    case 1:

        System.out.print("Enter course name: ");

        String course = sc.nextLine();

        courses.add(course);

        System.out.println("Course added successfully!");

        break;

    case 2:

        System.out.print("Enter course name to remove: ");

        String removeCourse = sc.nextLine();

        if (courses.remove(removeCourse)) {

            System.out.println("Course removed successfully!");

        } else {

            System.out.println("Course not found!");

        }

        break;

    case 3:

        StringBuffer sb = new StringBuffer();

        sb.append("\n----- Registered Courses -----\\n");

        if (courses.isEmpty()) {

            sb.append("No courses registered.\\n");

        } else {

            for (int i = 0; i < courses.size(); i++) {

                sb.append(i + 1)
```

```

        .append(" ")
        .append(courses.get(i))
        .append("\n");
    }
}

System.out.println(sb);

break;

case 4:

    System.out.println("Exiting...");

    break;

default:

    System.out.println("Invalid choice!");

}

} while (choice != 4);

sc.close();

}

}

```

Output

Output

```

Enter the number of tasks: 3
Enter Task 1: Java Assignment submission
Enter Task 2: OS test studies
Enter Task 3: Evening walk

----- TO-DO LIST -----
1. Java Assignment submission
2. OS test studies
3. Evening walk

```

Output

```
===== Student Course Registration System =====
1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 1
Enter course name: JAVA
Course added successfully!

===== Student Course Registration System =====
1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 1
Enter course name: Operating Systems
Course added successfully!

===== Student Course Registration System =====
1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 1
Enter course name: IKS
Course added successfully!

===== Student Course Registration System =====
1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 3
```

```
----- Registered Courses -----
1. JAVA
2. Operating Systems
3. IKS

===== Student Course Registration System =====
1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit|
Enter your choice: 2
Enter course name to remove: IKS
Course removed successfully!

===== Student Course Registration System =====
1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 3

----- Registered Courses -----
1. JAVA
2. Operating Systems

===== Student Course Registration System =====
1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 4
Exiting...
```

Conclusion

Through this assignment, we implemented dynamic data storage using `ArrayList` and `Vector`, both of which allow elements to be added, removed, and accessed without a fixed size constraint unlike traditional arrays, with the key distinction that `Vector` is synchronized and thread-safe while `ArrayList` is not. `StringBuffer` was also explored as a mutable alternative to the `String` class, enabling efficient character-level string manipulation operations such as `append`, `insert`, `reverse`, and `delete` without creating new objects in memory each time. Overall, this assignment demonstrated how `ArrayList`, `Vector`, and `StringBuffer` provide flexible and efficient solutions for dynamic data handling and string manipulation in real-world Java applications.